

SIH26002: Smart Logistics & Accessibility Platform

Backend Progress & AI Integration Report — North-Eastern Region (NER)

BACKEND LEAD

Arpit (MakeWithArpit)

AI COLLABORATION

Omji (AI-01, 02, 03, 08)

STATUS

Phases 0 to 5 COMPLETE
(100%)

TEST PASS RATE

34 / 34 Tests OK (100%)

Summary: Core backend architecture and the end-to-end intelligence loop are complete and pushed to GitHub. All 6 stages—incident reporting, PostGIS spatial snapping, geospatial disruption risk engine, NetworkX dual pathfinding, route re-ranking with safety explanations, and condition-aware ETA estimation—are verified with **34 passing tests**.

1. Overall Phase Completion Status

Phase	Domain / Milestone	Status	Key Deliverables Completed
Phase 0	Backend Foundation	COMPLETE	Django 5.1, GeoDjango, PostGIS 3.6.2, JWT auth, custom User/Profile, unified API envelope.
Phase 1	Field Reports & Vision Stub	COMPLETE	IncidentReport model (PointField, photo upload), AI-08 CV stub, Field Officer permissions.
Phase 2	Road Graph & Disruption Risk	COMPLETE	District & Infrastructure models, AI-01 Rule-based Risk Engine, spatial snapping (ST_DWithin).
Phase 3	Risk-Aware Route Optimization	COMPLETE	NetworkX PostGIS graph builder, dual Dijkstra pathfinding (shortest vs safest), AI-03 route ranking.
Phase 4	Condition-Aware ETA & Tracking	COMPLETE	Vehicle telemetry with O(1) cached coords, LocationPing breadcrumbs, Trip lifecycle, AI-02 ETA.
Phase 5	End-to-End Pipeline Demo	COMPLETE	Command <code>demo_pipeline</code> & API <code>POST /api/v1/routes/simulate-pipeline/</code> .
Phase 6–12	Weather, Alerts, Offline Sync	UPCOMING	Open-Meteo weather sync, automated critical alerts, Last-Write-Wins offline sync, dashboard.

2. Operational Intelligence Loop (Demonstrated & Verified)

Step 1: Field Incident Ingestion — Field Officer submits geo-tagged photo incident report via `POST /api/v1/reports/incidents/`. AI CV contract (AI-08) classifies issue and severity.

Step 2: PostGIS Proximity Snap — Spatial snap service locates nearest road segment within 1km buffer (`ST_DWithin`) and links it.

Step 3: Disruption Risk Surge — Geospatial Disruption Risk Engine (AI-01) recalculates weighted factor score (0–100) combining landslide susceptibility, historical count, rainfall, weather, and incident reports. Risk surges from MEDIUM (49) to HIGH (69).

Step 4: Dynamic Graph Rebuilding — PostGIS road network converted to weighted NetworkX graph with risk-penalized edge costs:
`weight = length × (1 + risk/15)`.

Step 5: Route Re-ranking & Recommendation — Route Ranking Engine (AI-03) detects high risk on primary highway and automatically switches recommendation to safe bypass with safety explanation.

Step 6: Condition-Aware ETA Estimation — ETA Service (AI-02) adjusts travel times factoring in North-Eastern road classification

speeds (NH 50km/h, SH 40km/h, MDR 30km/h, Rural 20km/h), hazard delays, and weather.

3. AI / ML Integration Contracts (For Omji)

Stable, drop-in service contracts matching AI/ML requirements. Omji can plug in trained models without changing controllers or DB schemas:

Contract ID	AI Service Wrapper	Inputs Provided by Backend	Output Schema Received
AI-08	<code>PhotoAnalysisService</code> apps/reports/services/	Uploaded photo file, GPS coordinates, officer description.	<code>ai_issue_type</code> , <code>ai_severity</code> , <code>ai_confidence</code> (0.0–1.0).
AI-01	<code>RiskPredictionService</code> apps/routes/services/	Segment landslide/flood hazard ratings, historical count, rainfall (mm), weather warning, incident reports.	<code>risk_score</code> (0–100), <code>risk_level</code> (low/med/high), <code>disruption_prob</code> , <code>top_factors</code> .
AI-03	<code>RouteRankingService</code> apps/routes/services/	List of candidate routes (<code>RouteCandidate</code> with distance, base ETA, aggregate risk).	Ranked candidates, <code>recommended</code> boolean, human-readable <code>explanation</code> .
AI-02	<code>ETAEstimationService</code> apps/vehicles/services/	Route segments, road classification, segment risk scores, rainfall mm, vehicle live telemetry speed.	<code>base_eta_minutes</code> , <code>predicted_eta_minutes</code> , <code>expected_delay_minutes</code> , <code>delay_reasons</code> .

4. Full REST API Endpoints Reference (For Frontend Team)

Method	Endpoint	Access Role	Purpose & Key Parameters
POST	/api/v1/accounts/login/	Public	JWT auth: returns access & refresh tokens.
GET	/api/v1/accounts/profile/	Authenticated	Fetch current user profile and assigned role.
POST	/api/v1/reports/incidents/	Field Officer	Submit geo-tagged photo incident report with multipart form.
GET	/api/v1/reports/incidents/	Authenticated	List incident reports (officers see own; admins see all).
GET	/api/v1/routes/districts/	Authenticated	List districts with GeoJSON boundaries and accessibility scores.
GET	/api/v1/routes/infrastructure/	Authenticated	List road segments. Filter by risk, status, or proximity: <code>?lat=&lng=&radius_m=</code>
POST	/api/v1/routes/calculate/	Authenticated	Calculate & rank routes. Body: <code>origin_node</code> , <code>destination_node</code> (or lat/lng).
POST	/api/v1/routes/simulate-pipeline/	Authenticated	Demo API: Simulates incident → snap → risk → rerank → ETA delay in 1 call.
POST	/api/v1/vehicles/{id}/locations/	Authenticated	Ingest telemetry ping & atomically update vehicle's live coordinates.
GET	/api/v1/vehicles/{id}/location/latest/	Authenticated	O(1) polling: High-performance instant fetch of vehicle position.
POST	/api/v1/trips/{id}/recalculate-eta/	Authenticated	Recomputes trip ETA using latest road conditions and live telemetry.

5. Test Suite Verification (100% Pass Rate)

Test Module	Tests Count	Result	Coverage Details
apps.accounts	4 tests	PASS	JWT auth, user registration, profile signal, standardized error envelope.
apps.reports	5 tests	PASS	Photo uploads, PointField spatial storage, AI CV stub, officer role permissions.
apps.routes	18 tests	PASS	Districts, proximity queries, AI-01 risk scoring, NetworkX pathfinding, route ranking, 7 pipeline E2E tests.
apps.vehicles	7 tests	PASS	Vehicle CRUD, ping ingestion, atomic cache, trip lifecycle, AI-02 ETA estimation.
Total Backend Suite	34 tests	34 / 34 OK	100% passing across all domain applications.

GitHub Repository: Code committed and pushed to `main` at `https://github.com/MakeWithArpit/SIH26002_Backend` (Commit: `a762df5`).